Allomaternal care, life history and brain size evolution in mammals

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Supplementary Information

Content

1. Supplementary methods
- 1.1. Categories of ecological variables p. 2
- 1.2. Sources of phylogenetic relationships p. 2
- 1.3. Sources of social behaviour, breeding system, ecology and life history p. 3

2. Supplementary results
- 2.1. Results of the Phylogenetic Principal Components Analyses p. 5
- 2.2. Results of an alternative coding scheme of helping behaviors p. 7
- 2.3. Analyses of the three major clades of primates p. 9

3. Supplementary discussion
- 3.1. Costs of gestation and lactation in mammals p. 10
- 3.2. Exclusion of bats and cetaceans p. 10

4. References p. 11

1. Supplementary methods

1.1. Categories of ecological variables

All groups were parametrized empirically (Table S1) to maximize the proportion of brain size variation that is explained by the respective variable, within the groups analysed (fissiped Carnivora, Rodentia, Primates and Artiodactyla). Here, we do not aim to explain or test correlations between brain size and ecological variables, but simply include them as covariates in our analyses to preclude the possibility that our results could be spurious due to some hidden effect.

Table S1: Categories and parametrization of ecological covariates

Diet	
aquatic faunivore, piscivore	1
frugivore/folivore, granivore	2
frugivore/faunivore, omnivore	3
herbivore, folivore	4
carnivore, faunivore, insectivore	5
Activity pattern	
cathemeral, crepuscular, nocturnal	1
diurnal	2
Substrate use	
aquatic, semi-aquatic	1
fossorial, semi-fossorial	2
terrestrial, semi-arboreal	3
arboreal	4

1.2. Sources of phylogenetic relationships

Phylogenetic relationships were taken mainly from Bininda-Emonds et al. (2007), supplemented by data from other studies to resolve polytomies and reflect more recent advances:

Primates:

Following Arnold et al. (2010) and Perelman et al. (2011)

Rodentia:

Cavioidea genera: Blanga-Kanfi et al. 2009

Muridae subfamilies: Steppan et al. 2004

Murinae genera: Lecompte et al. 2008

Arvicolinae genera: Buzan et al. 2008

Gerbillinae species: Chevret and Dobigny 2005

Rattus species: Rowe et al. 2011

Peromyscus species: Bradley et al. 2007

Microtus species: Jaarola et al. 2004

Marmotini genera and species: Herron et al. 2004

Cetartiodactyla:

Antilopinae genera: Agnarsson and May-Collado 2008

Afrotheria:

Hyracoidea next to Proboscidea: Kuntner et al. 2011

Carnivora:

Following Agnarsson et al. 2010:

Ursidae close to Pinnipedia

Ailurus close to Canidae

Viverridae position and *Nandinia* basal

Canidae species

Mustelidae genera

Herpestinae genera

Felinae genera

Ursus species

following Bininda-Emonds et al. 2007 and Eizirik et al. 2010:

Mephitidae not close to Procyonidae, and *Potos flavus* remaining with Procyonidae

Phocidae: Timing of *Phoca/Pusa* and *Halichoerus* from Fulton and Strobeck (2010).

Eulipotyphla:

Sorex species: groups according to Wilson and Reeder (2005) (*S. palustris* in the *vagrans* group, *S. cinereus* in own group)

Xenarthra

genera: Delsuc et al. 2002

1.3. Sources of social behaviour, breeding system, ecology and life history

Species numbers of the Excel data file are given in parentheses. In case of conflicting information, preference was given to sources from studies in the wild, and for secondary sources, more recent compilations were preferred over earlier ones.

Mammals in general, various species:

Bronson 1989; Clutton-Brock 2006; Emlen 1991; Gubernick 1981; Hayssen 1969-2009; Hayssen et al. 1993; Jennions and MacDonald 1994; Kleiman and Malcolm 1981; König 2006; Lewis and Pusey 1997; MacDonald 2001; Myers et al. 2006; Nowak 1999; Packer et al. 1992; Riedman 1982; Roulin 2002; Silk 2007; Snowdon 1996; Solomon and French 1997; Spencer-Booth 1970; Whitten 1987; Woodroffe and Vincent 1994

Rodentia

Dewsbury 1985; Hayes 2000; McGuire and Bemis 2007

Blumstein and Armitage 1998 (419-424, 431-436)

Blumstein and Armitage 1999 (422-424)

Carnivora

Creel and Creel 1991; Gittleman 1994; MacDonald et al. 2004; Wilson and Mittermeier 2009

Canidae: Malcolm 1985; Moehlman and Hofer 1997

Mustelidae: Johnson et al. 2000

Primates

Hrdy 1976; Kohda 1985; Mitani and Watts 1997; Nicolson 1987; Ross 2001; Ross 2003; Ross and MacLarnon 2000; Whitten 1987

Cercopithecines: Melnick and Pearl 1987

Colobines: McKenna 1979

Tarsiers: Gursky 2007

Platyrrhines: Goldizen 1987; Kleiman 1985; Santos et al. 1997; Tardif 1997

Lemurs: Garbutt 1999

Lagomorpha: Swihart 1984

'Insectivora': Symonds 1999

Single species:

Bales et al. 2000 (257)
Barry and Mundy 1998 (168)
Bernstein 1968 (298)
Bourlière et al. 1970 (271)
Burt and Grossenheider 1976 (405)
Calegaro-Marques and Bicca-Marques 1993 (238)
Carter and Roberts 1997 (381)
Clarke et al. 1998 (240)
Dixson and Fleming 1981 (237)
Ebensperger et al. 2002 (344)
Eberle and Kappeler 2006 (314)
Ekvall 1998 (51)
Fietz and Dausmann 2003 (313)
Fitch-Snyder and Ehrlich 2003 (331)
Frederick 1998 (332)
Hill 1964 (291)
Hill 1972 (297)
Hoogland 1995 (419)
Jolly 1966 (318)
Kawalika 2004 (333,334)
Klopfer 1981 (184)
Knöthig 2005 (232)
Lee 1987 (229)
Leighton 1987 (309,310)
MacDonald 1979 (86)
Manning et al. 1995 (392)
Mech et al. 1999 (70)
Miranda et al. 2005 (239)
Nekaris 2003 (321,322, 327, 328, 329)
Overdorff and Tecot 2006 (321)
Owens and Owens 1979a, b, 1996 (156)
Patris and Baudoin 2000 (392)
Perry et al. 1998 (215)
Porter 2001 (247)
Powell and Fried 1992 (382)
Pusey and Packer 1994 (140)
Rajanathan and Bennett 1990 (292)
Rood 1974 (152)
Rood 1978 (148)
Roonwal and Mohnot 1977 (302)
Rotundo et al. 2005 (235)
Ruffer 1965 (405)
Ruhiyat 1983 (294)
Sjare and Stirling 1996 (203)
Tarasov 1959 (101)
Thorington and Ferrell 2006 (430, 432)
Viljoen 1975 (425)
Wagner et al. 2008 (155)
Wheatley 1997 (347)
Whitten 1984 (275)
Wiens 2002 (329)

2. Supplementary results

2.1. Results of the Phylogenetic Principal Components Analysis

Taking phylogeny into account in the Principal Components Analysis results in two different components (Table S2a), the first one comprising communal nesting/babysitting and protection (pPC1), and the second one loading mainly on provisioning (pPC2). However, the very low Eigenvalues indicate that the original variables are already largely orthogonal to each other if phylogeny is taken into account. Therefore, rather than combing variables into factors, the phylogenetic PCA only selects those two variables that explain most of the variation, namely communal nesting/babysitting, and provisioning by the male. The same difficulty is encountered if phylogenetic PCA is applied to each other the large groups separately (Table S2b). Nevertheless, in order to corroborate the results from species-level PCA in the main text, we report here the full results for the PGLS regression using components from phylogenetic PCA (Table S3).

Table S2: Phyogenetic principal components analysis of helping behaviors in placental mammals

a) in placental mammals

Phylogenetic PCA			**Rotated factor loadings**						
						Provision		**Carry**	
	Eigen-value	**% Var. expl.**	**Allo-nursing**	**Comm. nesting / babysit/ retrieval**	**Protect**	**Male**	**Others**	**Male**	**Others**
pPC1	0.760	45.8	0.134	**0.943**	**0.770**	0.060	0.196	0.233	0.126
pPC2	0.417	25.1	0.082	0.017	0.213	**0.980**	**0.415**	0.119	-0.004

b) within large orders

Phylogenetic PCA						**Rotated factor loadings**				
							Provision		**Carry**	
Group		**Eigen-value**	**% Var. expl.**	**Allo-nursing**	**Comm. nesting/ babysit/ retrieval**	**Protect**	**Male**	**Others**	**Male**	**Others**
Carnivora	pPC1	1.303	**52.1**	0.107	0.017	0.211	**0.994**	0.278		
	pPC2	0.692	**27.7**	-0.292	**-0.941**	**-0.789**	-0.078	-0.554		
Rodentia	pPC1	**0.882**	**74.7**	0.272	**0.953**	0.288	0.108	0.064		
	pPC2	**0.244**	**20.7**	-0.032	0.303	**0.958**	0.14	0.099		
Primates	pPC1	**0.742**	**55.7**	0.015	**0.992**	0.057	0.383	0.263	**0.543**	0.35
	pPC2	**0.287**	**21.5**	-0.092	0.055	**0.979**	**0.432**	0.279	0.239	-0.126
Artiodactyla	pPC1	**0.432**	**83.5**	-0.056	0.127	**0.997**				
	pPC2	**0.080**	**15.4**	**0.693**	**0.990**	0.079				

Table S3: Phylogenetic multivariate regression models (PGLS) of the effects of the phylogenetic components of help within large orders, A) on brain size, B) on fertility, and C) on gestation, lactation, age at first reproduction, and maximum lifespan. Note that the nature of help summarized by pPC1 and pPC2 differs for the various orders, see Table S2b. For help factors and covariates, the table lists p-values followed by the direction of the effect (+ or -). Lambda values are estimated by Maximum Likelihood, the number of asterisks indicates significant differences from 0 or 1 (***: significantly different from both 0 and 1, **: significantly different from 1, *: significantly different from 0).

A) brain size as response

Brain size					Help factors from pPCA		Covariates				
	N	lambda	adjR2	AICc	pPC1	pPC2	Grega-riousness	Diet	Activity	Substrate	body mass
Carnivora	101	0.953***	0.937	-63.6	**<0.0001+**	0.835+	0.364-	0.501-	0.594+	0.234+	<0.0001+
Rodentia	113	0.915***	0.937	-58.5	0.791+	**0.008+**	0.178-	0.116 -	0.837+	**0.044+**	<0.0001+
Primates	98	0.991*	0.879	-74.5	0.662-	**0.010-**	0.090+	0.238 -	0.394+	0.239 -	<0.0001+
Artiodactyla	66	0.989*	0.902	-17.0	0.735-	0.302+	0.065+	0.390+	0.374-		<0.0001+

B) fertility as response

Fertility					Help factors from pPCA		Covariates				
	N	lambda	adjR2	AICc	pPC1	pPC2	Grega-riousness	Diet	Activity	Substrate	body mass
Carnivora	98	0.837***	0.129	138.2	0.287-	0.765-	0.304+	0.851+	0.382+	0.470+	<0.0001-
Rodentia	99	0.898***	0.134	141.1	0.423+	0.082-	0.325+	0.454 -	0.387 +	0.390 -	0.0004-
Primates	84	0.692**	0.589	89.6	**0.004+**	0.365+	0.757-	0.727 -	0.306 +	0.129 -	<0.0001-
Artiodactyla	60	0.855*	0.123	62.7	0.479-	0.667+	0.208-	0.238+	0.074+		0.006-

C) other life history variables as response

						Help factors from pPCA		Covariates				
Group	LH trait	N	lambda	adjR2	AICc	pPC1	pPC2	Grega-riousness	Diet	Activity	Substrate	body mass
Carnivora	gestation	97	0.975*	0.179	-27.8	0.333-	0.395+	0.345+	0.153-	0.136+	0.935+	<0.0001+
	lactation	90	0.901***	0.355	95.8	0.996-	0.134-	0.097+	0.639+	0.587+	0.794+	<0.0001+
	AFR	91	0.768***	0.436	86.2	0.250+	0.301-	0.264+	0.232+	0.758+	0.936-	<0.0001+
	max Lifespan	93	1*	0.268	-0.28	0.450+	0.436+	0.539+	**0.027-**	0.107+	0.052+	<0.0001+
Rodentia	gestation	105	1*	0.202	-45.4	0.143+	0.400+	0.374-	0.769+	0.287-	0.669+	<0.0001+
	lactation	103	0.821***	0.234	31.3	0.479+	**0.030+**	0.123-	0.259-	**0.017+**	**0.022+**	<0.0001+
	AFR	103	0.637***	0.328	170.8	0.094-	0.170+	0.539-	0.890-	0.596+	0.296+	<0.0001+
	max Lifespan	94	0.812***	0.215	109	0.676+	0.246+	0.664+	0.743-	**0.038-**	0.169+	<0.0001+
Primates	gestation	84	0.982***	0.28	-113.6	**0.040-**	0.585-	0.366+	0.437+	0.244+	0.875-	0.0001+
	lactation	85	0.746***	0.647	84.6	**0.016-**	0.383-	0.757-	0.353+	0.498+	0.379+	<0.0001+
	AFR	84	1*	0.101	49.7	0.086-	0.682+	0.907+	0.637+	0.409+	0.919-	0.006+
	max Lifespan	87	0.861***	0.293	-21.1	0.116-	0.509+	0.779-	0.996-	0.566+	0.777+	<0.0001+
Artiodactyla	gestation	65	1*	0.420	-58.3	**0.0002+**	0.874-	0.873+	0.324-	0.592+		<0.0001+
	lactation	52	0.428**	0.503	43.3	0.297+	0.179-	0.891+	0.258+	0.182+		<0.0001+
	AFR	60	0**	0.600	57.4	0.532+	0.433-	0.008+	0.054-	0.883+		<0.0001+
	max Lifespan	64	0.482**	0.400	-14.7	0.966-	0.084+	0.112+	0.499-	0.223-		<0.0001+

In comparison with the non-phylogenetic PCA reported in the main text, roughly similar factors emerge, although allonursing does usually not stand alone, but groups with other helping behaviours. In Carnivora, provisioning by the male (pPC2) is separated from protection and babysitting/retrieval (pPC1), which are combined into PC1 in the raw PCA. In Rodentia, communal nesting (pPC1) and protection (pPC2) are separated, whereas they were grouped together in PC2 in the raw model. The rare, but for our purposes interesting behaviour of provisioning is not represented in the phylogenetic factors. Likewise, allonursing does not appear in the first two phylogenetic components in Primates, but babysitting dominates pPC1, and protection dominates pPC2. Provisioning and carrying are somewhat distributed over these two factors, which is not very satisfying for interpretations of potential correlations with brain size or life history parameters. This weakness of resolution the phylogenetic PCA is easily explained by the fact that direct forms of help are phylogenetically clumped in primates, and thus do not have much of an effect in a phylogenetic PCA. In Artiodactyla, the phylogenetic factors load on the same parameters as the raw factors, but in reverse order. In conclusion, the phylogenetic factors of Carnivora and of Rodentia may be most helpful for further insights in the following multivariate models.

The phylogenetic PCA shows that in Carnivora, it is male provisioning that is responsible for the positive correlation with brain size, while babysitting/retrieval and protection by the male are not correlated with brain size at all. In Rodentia, on the other hand, where the second factor is split in the phylogenetic PCA, only protection by the male is positively correlated with brain size, whereas communal nesting is not. Likewise, in primates, male provisioning and protection are negatively correlated with brain size, whereas babysitting and carrying by the male are not. The results for Artiodactyla are confirmed by phylogenetic PCA.

With regard to fertility, the positive correlation with help is confirmed by phylogenetic PCA in Primates, but not the positive trend with allonursing in Carnivora. The correlation between a more altricial state of the neonates and the amount of help in Primates is also confirmed by phylogenetic PCA.

2.2. Results of an alternative coding scheme of helping behaviors

To investigate whether the results reported in this study are robust with respect to different coding schemes of helping behaviors, we conducted analogous tests with an alternative scheme. In this, the variable “Help1” is coded as 1 if provisioning by either the male or other group members was regularly observed, and 0 otherwise. The variable “Help2” is coded as 1 if if any babysitting, allonursing or provisioning behavior was observed, and 0 otherwise. Thus, Help1 is most closely reflecting classical “cooperatively breeding” species such as canid carnivores and callithrichid primates, and Help2 is a broader category including direct and indirect help (but not only protection alone). The results of PGLS regression models using this scheme are reported in Table S4. Although p-values vary slighly in comparison with Tables 2C), 3C), and 4) in the main text, on the whole, the results for brain size and fertility are strikingly similar. Therefore, we conclude that our results are robust against changes in the coding scheme, especially with respect to the frequencies of a behavior. Nevertheless, the more detailed original coding scheme yields more consistent results for models predicting other life history variables.

Table S4: Phylogenetic multivariate regression models (PGLS) within large orders of the effects of help, as estimated using alternative, categorical classifications, A) on brain size, B) on fertility, and C) on gestation, lactation, age at first reproduction, and maximum lifespan. For help factors and covariates, the table lists p-values followed by the direction of the effect (+ or -). Lambda values are estimated by Maximum Likelihood, the number of asterisks indicates significant differences from 0 or 1 (***: significantly different from both 0 and 1, **: significantly different from 1, *: significantly different from 0).

A) brain size as response

Brain size					**Help categories**		**Covariates**				
	N	**lambda**	**adjR2**	**AICc**	**Help1**	**Help2**	**Grega-riousness**	**Diet**	**Activity**	**Substrate**	**body mass**
Carnivora	101	0.844***	0.935	-56.8	**0.006+**	**0.029+**	0.348-	0.174-	0.620+	0.494+	<0.0001+
Rodentia	113	0.849***	0.935	-58.3	**0.031+**	0.507+	0.718+	0.129-	0.799+	**0.035+**	<0.0001+
Primates	98	0.985*	0.888	-84.3	**0.0003-**	0.674-	0.194+	0.263-	**0.023+**	0.190-	<0.0001+
Artiodactyla	66	0.990*	0.904	-19.8	---	0.242+	0.100+	0.324+	0.370-	---	<0.0001+

B) fertility as response

Fertility					**Help categories**		**Covariates**				
	N	**lambda**	**adjR2**	**AICc**	**Help1**	**Help2**	**Grega-riousness**	**Diet**	**Activity**	**Substrate**	**body mass**
Carnivora	98	0.631***	0.206	129.0	0.059+	0.095-	0.479+	0.825-	0.130+	0.368+	<0.0001-
Rodentia	99	0.812***	0.113	141.2	0.917-	0.311+	0.648-	0.387-	0.332+	0.459-	0.0004-
Primates	84	0**	0.784	84.0	**0.001+**	0.241+	**0.005+**	0.831-	0.150-	0.198-	<0.0001-
Artiodactyla	60	0.876*	0.131	60.8	---	0.781-	0.274-	0.257+	0.067+	---	0.008-

C) other life history variables as response

						Help categories		**Covariates**				
Group	**LH trait**	**N**	**lambda**	**adjR2**	**AICc**	**Help1**	**Help2**	**Grega-riousness**	**Diet**	**Activity**	**Substrate**	**body mass**
Carnivora	gestation	97	1*	0.157	-31.7	0.717-	**0.008+**	0.430+	0.201-	0.917-	0.959-	0.0001+
	lactation	90	0.775***	0.279	111.1	0.684-	0.996-	0.284+	0.487+	0.974+	0.939+	<0.0001+
	AFR	91	0.609***	0.426	84.5	0.223+	0.306+	0.714+	0.277+	0.850-	0.658-	<0.0001+
	max Lifespan	93	1*	0.249	-0.04	0.717+	0.955+	0.159+	**0.006-**	0.261+	0.055+	<0.0001+
Rodentia	gestation	105	1*	0.229	-51.3	**0.016+**	0.224+	0.608-	0.766+	0.382-	0.411+	<0.0001+
	lactation	103	0.231**	0.169	114.1	0.248+	0.686-	0.490-	0.732+	0.148+	**0.001+**	0.001+
	AFR	103	0.511***	0.306	170.8	0.531+	0.276-	0.945-	0.969-	0.644+	0.286+	<0.0001+
	max Lifespan	94	0.769***	0.206	106.0	0.290+	0.199+	0.682+	0.740-	**0.047-**	0.217+	0.001+
Primates	gestation	84	0.968***	0.253	-112.5	0.801-	0.095-	0.735+	0.515+	0.190+	0.769-	0.001+
	lactation	85	0.724*	0.585	87.2	0.069-	0.297-	0.788-	0.308+	0.122+	0.527+	<0.0001+
	AFR	84	1*	0.188	39.3	**0.003-**	0.158-	0.918+	0.454+	0.064+	0.530-	0.158+
	max Lifespan	87	0.742***	0.313	-23.8	0.070-	0.907+	0.764-	0.832-	0.223+	0.568+	0.001+
Artiodactyla	gestation	65	1*	0.285	-46.1	**---**	0.358+	0.668-	0.706+	0.640+	---	<0.0001+
	lactation	52	0.392**	0.496	43.2	---	0.586+	0.778-	0.192+	0.119+	---	<0.0001+
	AFR	60	0**	0.613	54.0	---	0.207+	**0.033**+	0.059-	0.524+	---	<0.0001+
	max Lifespan	64	0.768*	0.343	-15.7	---	0.188+	0.209+	0.692-	0.252-	---	<0.0001+

2.3. Analyses of the three major clades of primates

If the three large monophyletic groups of primates are analyzed separately (Table S5), we see that the trends within primates as agroup are largely determined by the trends within platyrrhines, the New World monkeys.

Table S5: Phylogenetic multivariate regression models (PGLS) of the phylogenetic components of help on brain size, fertility, and gestation, within the three major clades of primates. For help factors and covariates, the table lists p-values followed by the direction of the effect (+ or -).

				Help factors from PCA		Covariates				
	N	lambda	adjR2	PC1	PC2	Grega-rious-ness	Diet	Activity	Sub-strate	Body mass
All Primates										
brain size	98	0.989***	0.888	**0.010-**	**0.037+**	0.251+	0.231-	0.746+	0.432-	<0.001+
fertility	84	0.673**	0.602	**0.002+**	0.589-	0.080+	0.610-	0.867+	0.098-	<0.001-
gestation	84	0.982***	0.272	**0.037-**	0.642-	0.928+	0.411+	0.138+	0.915-	0.001+
Platyrrhini										
brain size	33	0.998*	0.914	**0.003-**	0.067+	0.069+	0.111+	0.285-		<0.0001+
fertility	28	0	0.867	**0.010+**	0.710-	0.053+	0.057-	0.230-		0.004-
gestation	28	0.927**	0.789	**0.005-**	0.367-	**0.026+**	0.120-	0.613+		0.893+
Catarrhini										
brain size	43	0.971***	0.836	0.280-	0.475+	0.721+	0.151-		0.159-	<0.001+
fertility	37	0**	0.551	0.994+	0.453+	0.275+	0.733+		0.300-	0.005-
gestation	35	0.853**	0.109	0.518-	0.787+	0.536-	0.734+		0.827+	0.067+
Strepsirrhini										
brain size	20	0.820**	0.959	0.115+	0.285-	0.798+	0.280-	0.727-		<0.0001+
fertility	17	0**	0.459	0.133+	0.799+	0.473+	0.560-	0.413-		0.082-
gestation	19	0.993*	0.188	0.629+	0.776-	0.813+	0.648+	0.609-		0.020+

3. Supplementary discussion

3.1. Allomaternal care in bats and cetaceans

We did not compile data for bats and cetaceans because reliable data on allomaternal care and life history characteristics of both cetaceans and bats are notoriously difficult to obtain. Moreover, brain size in these two very groups may underlie different constraints than in other mammals (cf. Marino 2004, Marino et al. 2006, Dechmann and Safi 2009), making it more conservative to leave them out a priori.

Nevertheless, adding a few species from these groups would not change our results for the relation between allomaternal care and brain size in mammals as a group, because the available data suggests that cetaceans have large brains and probably some allomaternal care, whereas bats have small brains and almost exclusive maternal care, supporting the general relationship between brain size and allomaternal care we found in other mammals. Odontocete cetaceans are among the largest-brained mammals relative to their body mass, and some species exhibit allomaternal care (mainly babysitting, e.g. in sperm whales, killer whales or bottlenose dolphins, and, rarely, allonursing; Leung 2010, Mann et al. 2000; Whitehead 1996; Perrin et al. 2008). The main problem is the lack of evidence for the absence of such behaviors in other cetaceans. Bats, on the other hand, are relatively small-brained, and in most species mothers very likely care for their offspring alone (reviewed in Crichton and Krutzsch 2000). A few species may exhibit accidental allonursing, and some paternal protection of the roost in some tropical bats, also without evidence for absence in related species.

3.2. Costs of gestation and lactation in mammals

The energy transfer from mother to offspring through the placenta has a higher level of efficiency compared to that through milk production. Therefore we would expect costs of lactation per unit mass growth to be higher during lactation than during gestation, especially because larger offspring also need energy to grow, to maintain their larger body and to support their activity.

Energy intake is usually considerably higher in lactating than in gestating females (reviewed in Speakman 2008). However, as depletion of energy stores should also be taken into account, costs of gestation and lactation should be measured either by comparing resting metabolic rates or daily energy expenditure to the non-breeding status (Gittleman and Thompson 1988). In humans, a woman needs about 13% more energy during late gestation, and about 18% more energy during peak lactation (12 weeks postpartum) than in her non-reproductive state (van Raaij et al. 1991, reviewed in Rogers et al. 1997). For coyotes (*Canis latrans*), the additional energy requirements are 6% during gestation, but 142% during lactation, and 50% during post-weaning pup dependence, for both males and females (Laundre and Hernandez 2003). However, differences in body mass should also be taken into account in calculating the additional costs of reproduction. Table S6 lists values of Resting Metabolic Rates (RMR) for small mammals during late gestation and peak lactation, showing the deviation in percent from the metabolic rate predicted by Kleiber's equation ($3.42*\text{body mass}^{0.75}$, following Cretegny and Genoud 2006). In conclusion, costs of lactation seem to be considerably higher in most, if not all mammals, as compared to costs of gestation.

Table S6: Resting metabolic rates during gestation and lactation, in precent of the values predicted by Kleiber for the respective body mass.

	% of Kleiber			% increase vs. BMR		
Species	Non-Reprod. (BMR)	Gestation (RMR_G)	Lactation (RMR_L)	Gestation	Lactation	Source
Monodelphis dom. (marsupial)	63.5	82.2	142.6	18.7	79.1	Harder et al. 1996
Plecotus auritus (bat)	73	76.5	115	3.5	42	McLean and Speakman 2000
Shrews						
Microgale cowani	137	166	189	29	52	Stephenson and Racey 1993
M. dobsoni	71	97	168	26	97	
M. talazaci	73	78	85	5	12	
Rodents						
Akodon azarae	98	170	203	72	105	Antinuchi and Busch 2001
Ctenomys talarum	78	84.3	110	6.3	32	Zenuto 2002
Heterocephalus glaber	90.9	180.5	228.4	89.6	137.5	Urison and Buffenstein 1999
Mesocricetus auratus	106	126	139	20	33	Garton et al. 1994

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